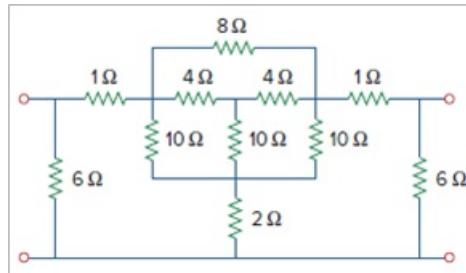


Problem

Use *PSpice* or *MultiSim* to obtain the z parameters of the network in Fig. 19.122.

Figure 19.122



Step-by-step solution

Step 1 of 5

Draw the circuit diagram as shown below.

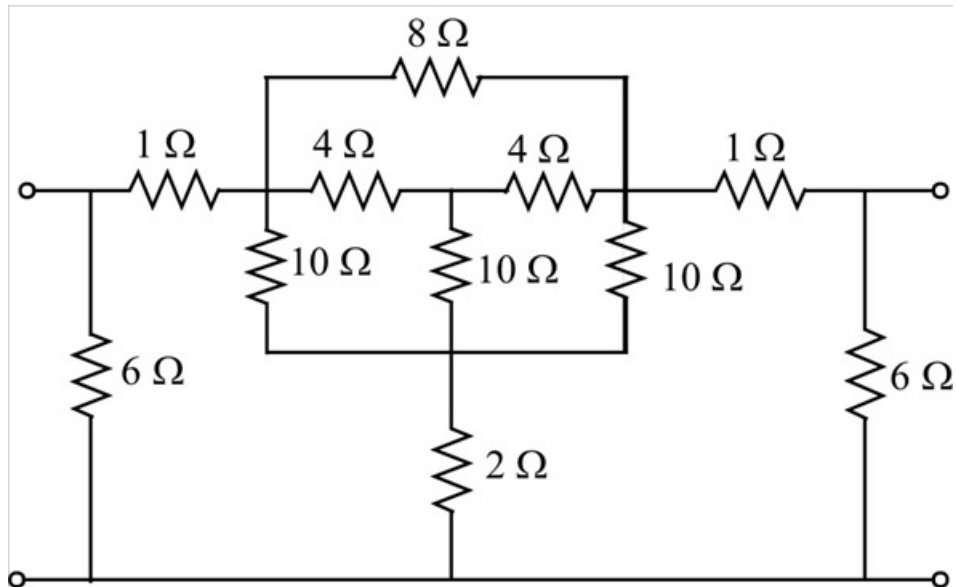


Figure 1

Step 2 of 5

Find z -parameters using Pspice.

Draw the Pspice to obtain z_{11} and z_{21} .

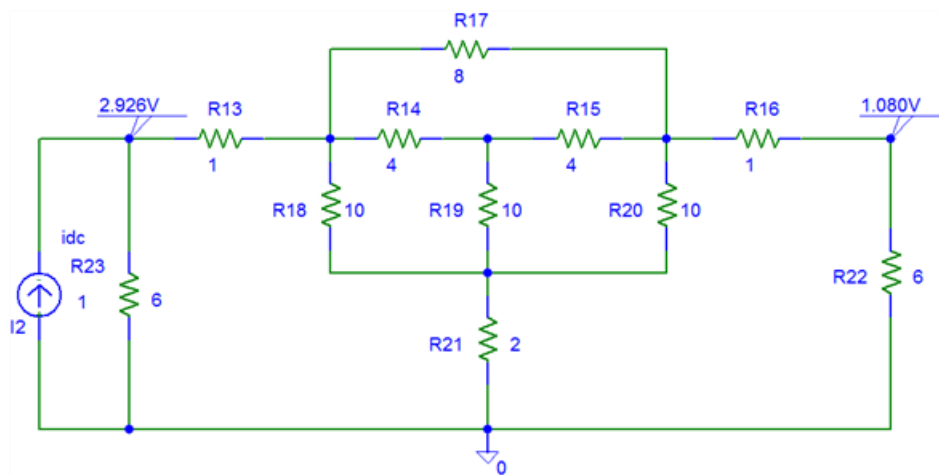


Figure 2

Step 3 of 5

Draw the Pspice to obtain z_{12} and z_{22} .

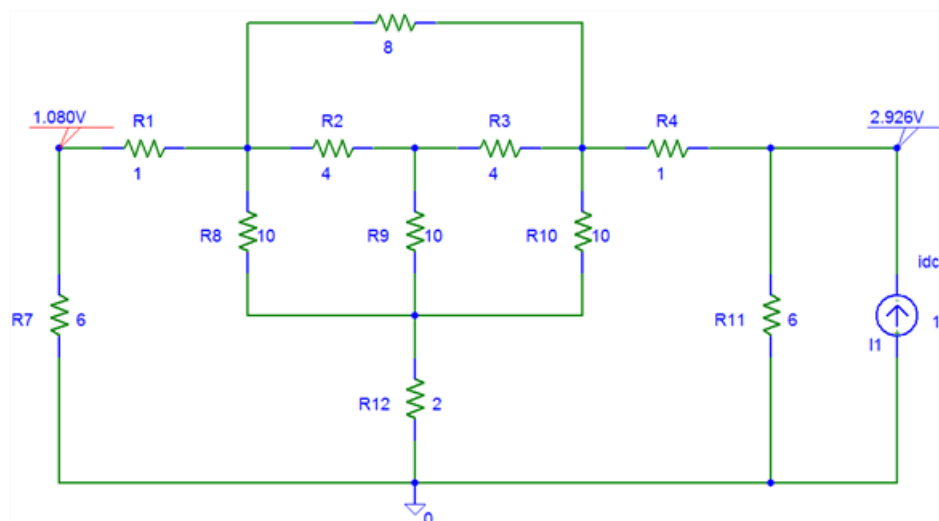


Figure 3

Step 4 of 5

From Figure 2, write z_{11} and z_{21} .

$$\begin{aligned}
 z_{11} &= \frac{V_1}{I_1} \\
 &= \frac{2.926}{1} \\
 &= 2.926 \, \Omega \\
 z_{21} &= \frac{V_2}{I_1} \\
 &= \frac{1.080}{1} \\
 &= 1.080 \, \Omega
 \end{aligned}$$

Step 5 of 5

From Figure 3, write z_{21} and z_{22} .

$$\begin{aligned} z_{12} &= \frac{V_1}{I_2} \\ &= \frac{1.080}{1} \\ &= 1.080 \, \Omega \end{aligned}$$

$$\begin{aligned} z_{22} &= \frac{V_2}{I_2} \\ &= \frac{2.926}{1} \\ &= 2.926 \, \Omega \end{aligned}$$

Therefore, the z -parameters are:

$$\boxed{[z] = \begin{bmatrix} 2.926 & 1.080 \\ 1.080 & 2.926 \end{bmatrix} \Omega} .$$